

Lodar Developer

Technical Engineering & AI Portfolio • Projects Showcase

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Autonomous LLM Workflow & Agent Engine

Python • Ollama • Qwen • FastAPI

Built an autonomous multi-agent pipeline leveraging local LLMs (Qwen/Llama) for automated data extraction, structured reasoning, and browser automation.

- Architected an event-driven agent loop with Playwright and local Ollama inference, achieving sub-second structured JSON extraction.
- Integrated dynamic DOM parsing and self-healing web form automation across 5+ enterprise platforms.

Real-Time Object Detection & Tracking for Autonomous Systems

Python • PyTorch • OpenCV • YOLOv8

Designed a low-latency real-time perception pipeline for obstacle detection and lane recognition in driving scenarios.

- Trained and fine-tuned customized YOLOv8 and TensorRT models on specialized vehicle/pedestrian datasets.
- Achieved 60+ FPS real-time video stream inference on edge GPU hardware with 94.2% mAP accuracy.

High-Throughput Distributed Microservices Platform

Go • Python • FastAPI • Docker

Engineered a scalable microservice architecture processing asynchronous event streams and transactional data.

- Implemented high-concurrency gRPC and REST communication layers backed by Redis caching and Kafka message queues.
- Designed resilient database schemas with PostgreSQL, query indexing, and automated connection pooling.

Cloud Data Analytics & Automated ETL Pipeline

Python • Pandas • SQL • Apache Spark

Constructed automated data pipelines extracting, cleaning, and transforming multi-source financial and telemetry datasets.

- Engineered automated ETL workflows using Apache Airflow, processing 500GB+ daily batch logs without data loss.
- Optimized analytical SQL queries and database partitions, accelerating dashboard query rendering by 4x.

Full-Stack Collaborative Workspace & Dashboard

TypeScript • React • Next.js • TailwindCSS

Developed a modern, responsive web application with real-time collaboration, interactive charts, and role-based access.

- Built responsive frontend UI with React, TypeScript, and modern component design tokens.
- Implemented real-time bidirectional messaging and state synchronization using WebSockets.

Embedded IoT Sensor Telemetry & Firmware

C++ • C • FreeRTOS • ESP32

Programmed bare-metal and RTOS firmware for microcontroller sensor nodes transmitting environmental and motion telemetry.

- Developed efficient low-power C++ firmware communicating over MQTT/TLS via Wi-Fi and Bluetooth BLE.
- Interfaced with multiple digital sensors using hardware I2C and SPI buses with interrupt-driven I/O.